

Презентация по лабораторной работе 5

Простые сети в GNS3. Анализ трафика

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1. Информация
2. Вводная часть
3. Выполнение заданий
4. Выводы

Раздел 1

1. Информация

1.1 Докладчик

► Седохин Даниил Алексеевич

1.1 Докладчик

- ▶ Седохин Даниил Алексеевич
- ▶ Российский университет дружбы народов им. П. Лумумбы

Раздел 2

2. Вводная часть

2.1 Цели и задачи

- ▶ Построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.

Раздел 3

3. Выполнение заданий

3.1 слайд 1

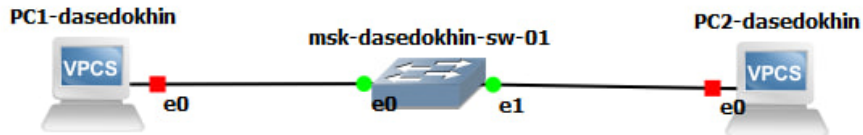
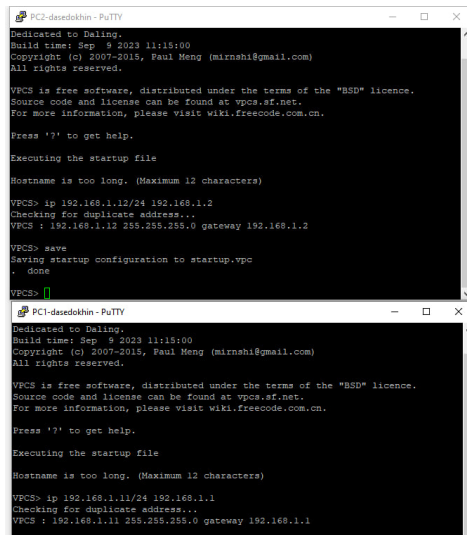


Рисунок 1: Воспроизведение топологии сети

3.2 слайд 2



The image shows two screenshots of a PuTTY terminal window titled "PC2-dasedokhin - PuTTY". The terminal displays the VPCS startup sequence, including copyright information and license details. The user enters the command "ip 192.168.1.12/24 192.168.1.2", and the system responds with "Checking for duplicate address..." and "VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.2". The user then enters "save", and the system responds with "Saving startup configuration to startup.vpc" and ". done". The prompt "VPCS>" is shown at the bottom.

```
PC2-dasedokhin - PuTTY
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.12/24 192.168.1.2
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.2

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS>
```

The second screenshot shows the same terminal window with the command "ip 192.168.1.11/24 192.168.1.1" entered. The system responds with "Checking for duplicate address..." and "VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1". The prompt "VPCS>" is shown at the bottom.

```
PC1-dasedokhin - PuTTY
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

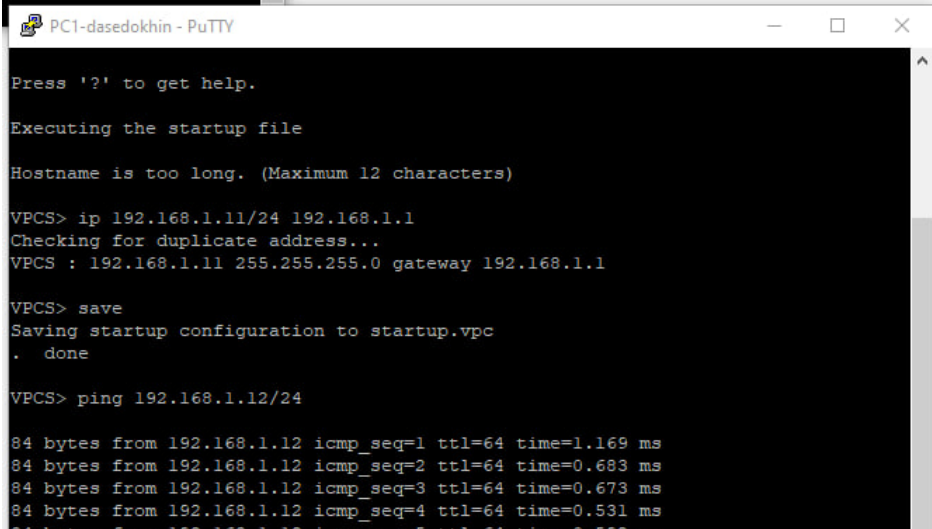
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

VPCS>
```

3.3 слайд 3



```
PC1-dasedokhin - PuTTY

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 192.168.1.12/24

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.169 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=0.683 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=0.673 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=0.531 ms
```

3.4 слайд 4

Закет из Standard input (PC1-dasedokhin Ethernet0 to msk-dasedokhin-sw-01 Ethernet0)

Файл Правка Вид Запуск Закет Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.006585	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.051212	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
4	0.056895	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.051947	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.050668	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.052292	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.059108	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)

▼ Frame 8: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

Section number: 1

► Interface id: 0 (-)

Encapsulation type: Ethernet (1)

Arrival Time: Dec 20, 2025 20:17:51.886227000 RTZ 2 (смм)

UTC Arrival Time: Dec 20, 2025 17:17:51.886227000 UTC

Epoch Arrival Time: 1766251071.886227000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.006816000 seconds]

[Time delta from previous displayed frame: 0.006816000 seconds]

[Time since reference or first frame: 2.059108000 seconds]

Frame Number: 8

Frame Length: 64 bytes (512 bits)

Capture Length: 64 bytes (512 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethiethertype:arp]

[Coloring Rule Name: ARP]

[Coloring Rule String: arp]

▼ Ethernet II, Src: Private 66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

► Destination: Broadcast (ff:ff:ff:ff:ff:ff)

► Source: Private 66:68:00 (00:50:79:66:68:00)

Type: ARP (0x0806)

[Stream Index: 3]

Padding: 00000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

▼ Address Resolution Protocol (request/gratuitous ARP)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0000)

Hardware size: 6

Protocol size: 4

3.5 слайд 5

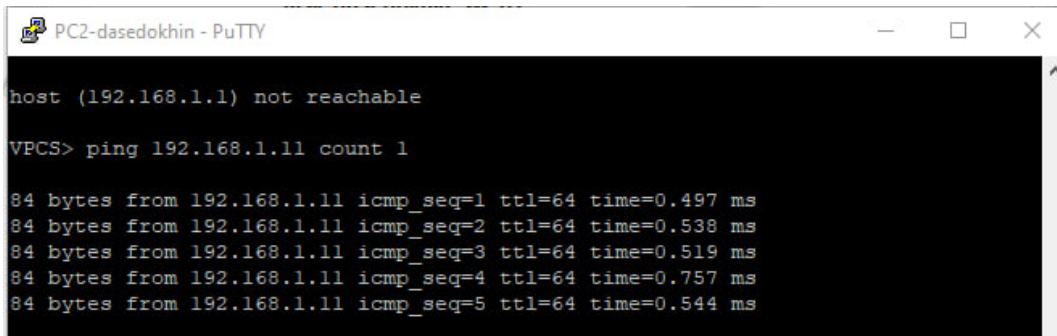
```
PC2-dasedokhin - PuTTY

VPCS> ping /?

ping HOST [OPTION ...]
  Ping the network HOST. HOST can be an ip address or name
  Options:
    -1          ICMP mode, default
    -2          UDP mode
    -3          TCP mode
    -c count   Packet count, default 5
    -D          Set the Don't Fragment bit
    -f FLAG    Tcp header FLAG |C|E|U|A|P|R|S|F|
                  bits |7 6 5 4 3 2 1 0|
    -i ms      Wait ms milliseconds between sending each packet
    -l size    Data size
    -P protocol Use IP protocol in ping packets
                  1 - ICMP (default), 17 - UDP, 6 - TCP
    -p port    Destination port
    -s port    Source port
    -T ttl     Set ttl, default 64
    -t          Send packets until interrupted by Ctrl+C
    -w ms      Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
```

3.6 слайд 6



```
PC2-dasedokhin - PuTTY

host (192.168.1.1) not reachable

VPCS> ping 192.168.1.11 count 1

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.497 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.538 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.519 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.757 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.544 ms
```

Рисунок 6: Запрос в ICMP моде к узлу PC1

3.7 слайд 7

Захват из Standard input (PCI-basedokhin Ethernet0) to msk-basedokhin-sw-01 Ethernet0

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... «Ctrl-F»

No.	Time	Source	Destination	Protocol	Length	Info
13	197.628741	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.12
14	198.628894	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.12
15	214.355821	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
16	214.355892	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
17	214.357413	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x14db, seq=1/256, ttl=64 (reply in 18)
18	214.357717	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x14db, seq=1/256, ttl=64 (request in 17)
19	215.359629	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x15db, seq=2/512, ttl=64 (reply in 20)
20	215.359908	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x15db, seq=2/512, ttl=64 (request in 19)
21	216.361481	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x16db, seq=3/768, ttl=64 (reply in 22)
22	216.361793	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x16db, seq=3/768, ttl=64 (request in 21)
23	217.363267	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x17db, seq=4/1024, ttl=64 (reply in 24)
24	217.363579	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x17db, seq=4/1024, ttl=64 (request in 23)
25	218.366557	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x18db, seq=5/1280, ttl=64 (reply in 26)
26	218.366842	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x18db, seq=5/1280, ttl=64 (request in 25)
27	259.351976	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x41db, seq=1/256, ttl=64 (reply in 28)
28	259.352057	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x41db, seq=1/256, ttl=64 (request in 27)
29	260.354413	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x42db, seq=2/512, ttl=64 (reply in 30)
30	260.354523	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x42db, seq=2/512, ttl=64 (request in 29)
31	261.359401	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x43db, seq=3/768, ttl=64 (reply in 32)

Frame 23: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Encapsulation type: Ethernet (1)

Arrival Time: Dec 20, 2025 20:21:27.190326000 RTZ 2 (sumo)

UTC Arrival Time: Dec 20, 2025 17:21:27.190326000 UTC

Epoch Arrival Time: 1766251287.190326000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 1.001414000 seconds]

[Time delta from previous displayed frame: 1.001414000 seconds]

[Time since reference or first frame: 217.363287000 seconds]

Frame Number: 23

Frame Length: 98 bytes (784 bits)

Capture Length: 98 bytes (784 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethertype:ip:icmp:data]

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

Source: Private_66:68:01 (00:50:79:66:68:01)

Type: IPv4 (0x0000)

[Stream index: 4]

Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0xbdb17 (56087)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

3.8 слайд 8

```
VPCS> ping 192.168.1.11 tcp=80 count=1
```

```
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.425 ms  
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.450 ms  
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.503 ms  
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.717 ms  
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.672 ms
```

```
VPCS> ping 192.168.1.11 tcp=80 count=1
```

```
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.417 ms  
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.577 ms  
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.572 ms  
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.502 ms  
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.493 ms
```


3.9 слайд 9

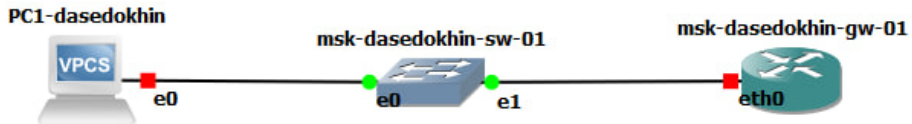


Рисунок 9: Воспроизведение топологии сети

3.10 слайд 10

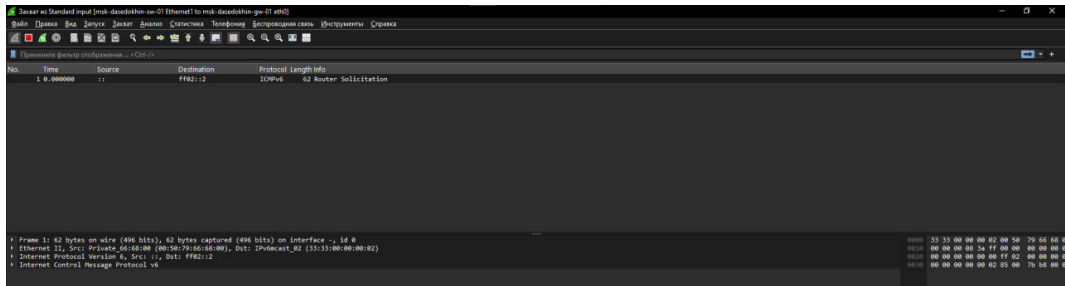
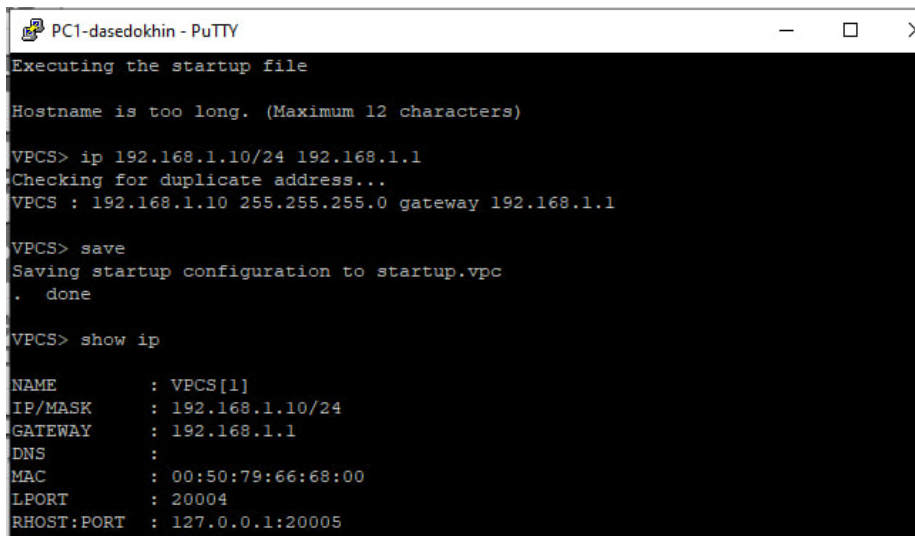


Рисунок 10: Включение захвата трафика между коммутатором и маршрутизатором

3.11 слайд 11



```
PC1-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

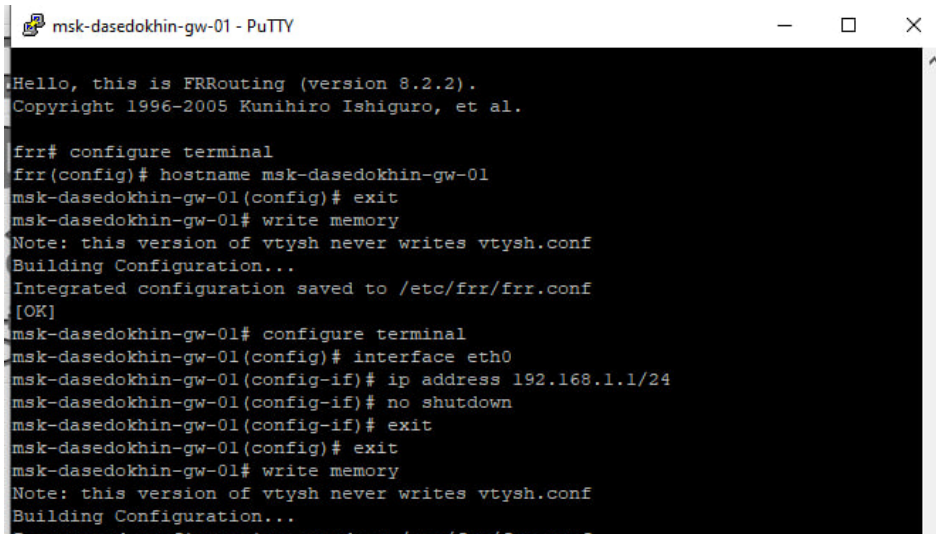
VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
```

3.12 слайд 12

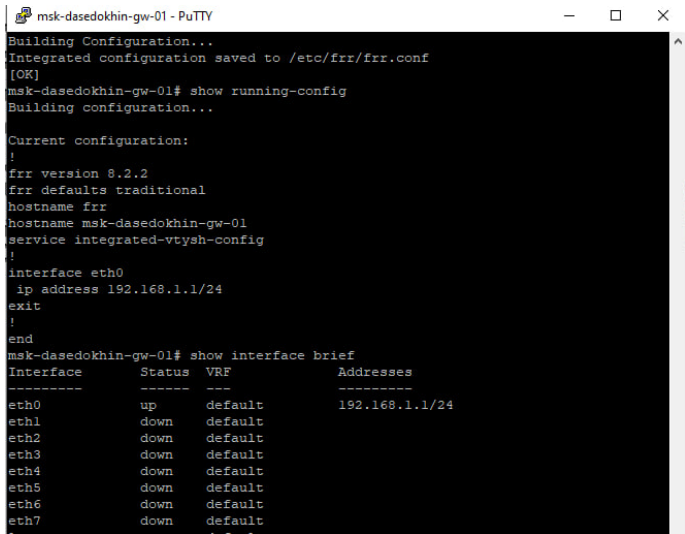


```
msk-dasedokhin-gw-01 - PuTTY

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-dasedokhin-gw-01
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01# configure terminal
msk-dasedokhin-gw-01(config)# interface eth0
msk-dasedokhin-gw-01(config-if)# ip address 192.168.1.1/24
msk-dasedokhin-gw-01(config-if)# no shutdown
msk-dasedokhin-gw-01(config-if)# exit
msk-dasedokhin-gw-01(config)# exit
msk-dasedokhin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
```

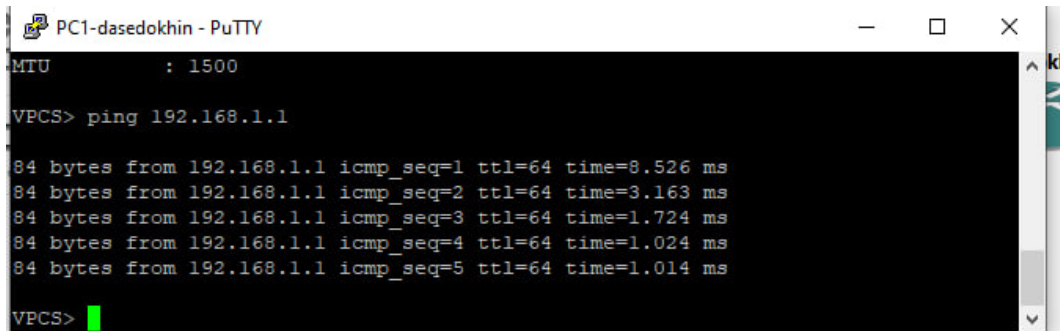
3.13 слайд 13



```
msk-dasedokhin-gw-01 - PuTTY
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dasedokhin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dasedokhin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-dasedokhin-gw-01# show interface brief
Interface      Status VRF      Addresses
-----
eth0           up    default  192.168.1.1/24
eth1           down  default
eth2           down  default
eth3           down  default
eth4           down  default
eth5           down  default
eth6           down  default
eth7           down  default
```

3.14 слайд 14



```
PC1-dasedokhin - PuTTY
MTU      : 1500

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=8.526 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=3.163 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.724 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.024 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.014 ms

VPCS> █
```

Рисунок 14: Проверка подключения

3.15 слайд 15

Захват из Standard input (msk-dasedokhin-gw-01 Ethernet1 to msk-dasedokhin-gw-01 eth0)

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
7	106.327934	::	ff02::1:ff23:0	ICMPv6	86	Neighbor Solicitation for fe80::e0c:edff:fe23:0
8	107.372923	fe80::e0c:edff:fe23::	ff02::1:ff23:0	ICMPv6	150	Multicast Listener Report Message v2
9	107.396031	fe80::e0c:edff:fe23::	ff02::1:ff23:0	ICMPv6	90	Multicast Listener Report Message v2
10	108.167112	fe80::e0c:edff:fe23::	ff02::1:ff23:0	ICMPv6	150	Multicast Listener Report Message v2
11	108.326053	fe80::e0c:edff:fe23::	ff02::1:ff23:0	ICMPv6	90	Multicast Listener Report Message v2
12	166.823400	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
13	166.826216	0c:0c:ed:23:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:0c:ed:23:00:00
14	166.827314	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xffdc, seq=1/256, ttl=64 (reply in 15)
15	166.835325	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xffdc, seq=1/256, ttl=64 (request in 14)
16	167.830383	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x00dd, seq=2/512, ttl=64 (reply in 17)
17	167.841080	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x00dd, seq=2/512, ttl=64 (request in 16)
18	168.842041	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x01dd, seq=3/768, ttl=64 (reply in 19)
19	168.844067	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x01dd, seq=3/768, ttl=64 (request in 18)
20	169.847123	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x02dd, seq=4/1024, ttl=64 (reply in 21)
21	169.847795	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x02dd, seq=4/1024, ttl=64 (request in 20)
22	170.848540	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x03dd, seq=5/1280, ttl=64 (reply in 23)
23	170.849239	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x03dd, seq=5/1280, ttl=64 (request in 22)
24	171.844609	0c:0c:ed:23:00:00	Private_66:68:00	ARP	60	Who has 192.168.1.10? Tell 192.168.1.1
25	171.844921	Private_66:68:00	0c:0c:ed:23:00:00	ARP	60	192.168.1.10 is at 00:50:79:66:68:00

* Frame 22: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

- Section number: 1
- Interface id: 0 (-)
- Encapsulation type: Ethernet (1)
- Arrival Time: Dec 20, 2025 20:29:39.400661000 RTZ 2 (sma)
- UTC Arrival Time: Dec 20, 2025 17:29:39.400661000 UTC
- Epoch Arrival Time: 1766251779.400661000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 1.000745000 seconds]
- [Time delta from previous displayed frame: 1.000745000 seconds]
- [Time since reference or first frame: 170.848540000 seconds]
- Frame Number: 22
- Frame Length: 98 bytes (784 bits)
- Capture Length: 98 bytes (784 bits)
- [Frame is marked: false]
- [Frame is ignored: false]
- [Protocols in frame: ethiethertype:ipicmp:data]
- [Coloring Rule Name: ICMP]
- [Coloring Rule String: icmp || icmpv6]
- * Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:0c:ed:23:00:00 (0c:0c:ed:23:00:00)
 - Destination: 0c:0c:ed:23:00:00 (0c:0c:ed:23:00:00)
 - Source: Private_66:68:00 (00:50:79:66:68:00)
 - Type: IPv4 (0x0800)
 - [Stream index: 4]
- * Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 64
 - Identification: 0x0003 (56579)
 - 0000..... = Flags: 0x00

3.16 слайд 16

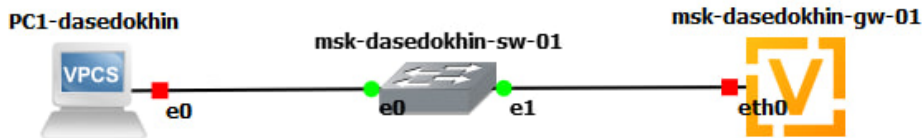
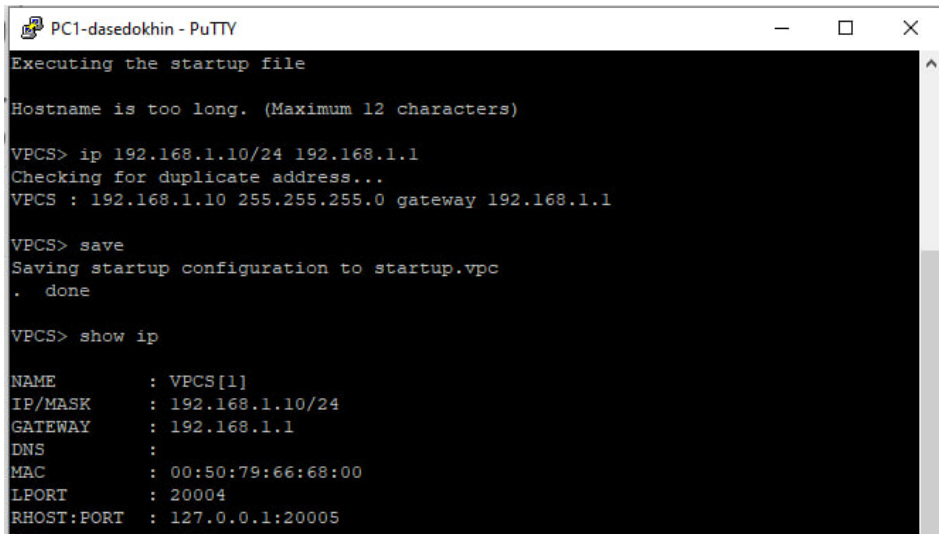


Рисунок 16: Воспроизведение топологии сети

3.17 слайд 17



```
PC1-dasedokhin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 20004
RHOST:PORT : 127.0.0.1:20005
```

3.18 слайд 18

```
mik-dsiedokhin-gw-01 - PuTTY
[ 16.616485] bridge: filtering via arp/ip/ip6tables is no longer available by
default. Update your scripts to load br_netfilter if you need this.
[ 16.624771] random: systemd: uninitialized urandom read (16 bytes read)
[ 16.673802] Bridge firewalling registered
[ 16.761556] systemd[1]: Started VyOS configuration daemon.
[ 16.882578] mpls_gso: MPLS GSO support
[ 17.118962] systemd[1]: Started Load Kernel Modules.
[ 17.135375] systemd[1]: Condition check resulted in Kernel Configuration File
System being skipped.
[ 17.206555] systemd[1]: Starting Apply Kernel Variables...
[ 17.216828] systemd[1]: Condition check resulted in FUSE Control File System
being skipped.
[ 17.242743] systemd[1]: Started Load/Save Random Seed.
[ 17.491062] systemd[1]: Started Create System Users.
[ 17.529918] random: systemd-journal: uninitialized urandom read (16 bytes rea
d)
[ 17.542117] systemd[1]: Starting Create Static Device Nodes in /dev...

Welcome to VyOS - vyos tty30

vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
Welcome to the VyOS install program. This script
will walk you through the process of installing the
VyOS image to a local hard drive.
Would you like to continue? (Yes/No) [Yes]: yes
Probing drives: OK
The VyOS image will require a minimum 2000MB root.
Would you like me to try to partition a drive automatically
or would you rather partition it manually with parted? If
you have already setup your partitions, you may skip this step

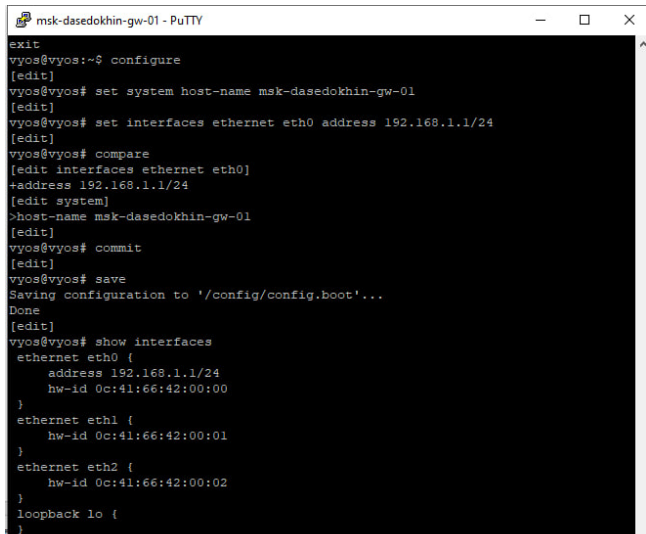
Partition (Auto/Parted/Skip) [Auto]: auto

I found the following drives on your system:
sda    8589MB
sdb     1MB

Install the image on? [sda]:sda

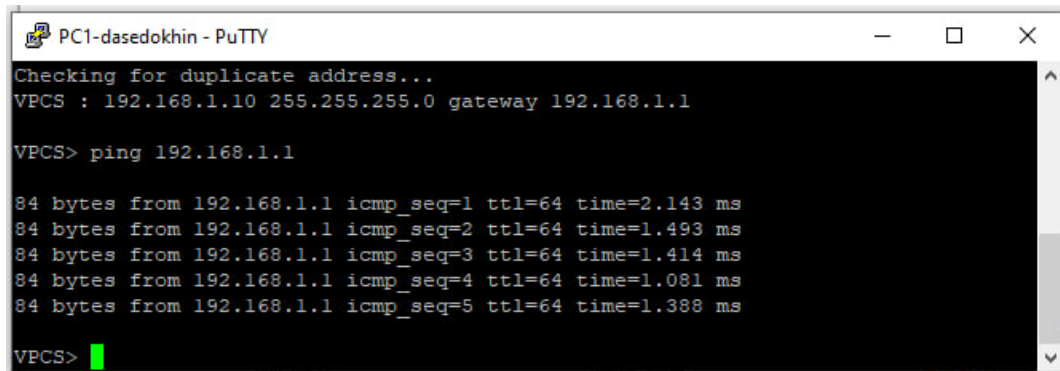
This will destroy all data on /dev/sda.
Continue? (Yes/No) [N]:n
```

3.19 слайд 19



```
msk-dasedokhin-gw-01 - PuTTY
exit
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dasedokhin-gw-01
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit system]
>host-name msk-dasedokhin-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:41:66:42:00:00
  }
  ethernet eth1 {
    hw-id 0c:41:66:42:00:01
  }
  ethernet eth2 {
    hw-id 0c:41:66:42:00:02
  }
  loopback lo {
  }
```

3.20 слайд 20



```
PC1-dasedokhin - PuTTY
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.143 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.493 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.414 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.081 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.388 ms

VPCS> 
```

Рисунок 20: Проверка подключения

3.21 слайд 21

Захват из Standard input [msk-dasedokhin-sw-01 Ethernet1 to msk-dasedokhin-sw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
14	515.513961	::	ff02::1:ff42:0	ICMPv6	86	Neighbor Solicitation for fe80::e41:66ff:fe42::
15	516.539497	fe80::e41:66ff:fe42::	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
16	516.541392	fe80::e41:66ff:fe42::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
17	517.057212	fe80::e41:66ff:fe42::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
18	517.435206	fe80::e41:66ff:fe42::	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
19	729.363120	Private 66:68:00	Broadcast	ARP	64	who has 192.168.1.1? Tell 192.168.1.10
20	729.366719	0c:41:66:42:00:00	Private 66:68:00	ARP	60	192.168.1.1 is at 0c:41:66:42:00:00
21	729.367959	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x31e3, seq=1/256, ttl=64 (reply in 22)
22	729.369392	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x31e3, seq=1/256, ttl=64 (request in 21)
23	730.374575	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x32e3, seq=2/512, ttl=64 (reply in 24)
24	730.375541	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x32e3, seq=2/512, ttl=64 (request in 23)
25	731.377614	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x33e3, seq=3/768, ttl=64 (reply in 26)
26	731.378759	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x33e3, seq=3/768, ttl=64 (request in 25)
27	732.381840	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x34e3, seq=4/1024, ttl=64 (reply in 28)
28	732.382763	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x34e3, seq=4/1024, ttl=64 (request in 27)
29	733.387687	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x35e3, seq=5/1280, ttl=64 (reply in 30)
30	733.388935	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x35e3, seq=5/1280, ttl=64 (request in 29)
31	734.582630	0c:41:66:42:00:00	Private 66:68:00	ARP	60	who has 192.168.1.10? Tell 192.168.1.1
32	734.583845	Private 66:68:00	0c:41:66:42:00:00	ARP	60	192.168.1.10 is at 00:50:79:66:68:00

Frame 29: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Section number: 1

- Interface id: 0 (-)
- Encapsulation type: Ethernet (1)
- Arrival Time: Dec 20, 2025 20:56:05.882474000 RTZ 2 (сумма)
- UTC Arrival Time: Dec 20, 2025 17:56:05.882474000 UTC
- Epoch Arrival Time: 1766253365.882474000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 1.004924000 seconds]
- [Time delta from previous displayed frame: 1.004924000 seconds]
- [Time since reference or first frame: 733.387687000 seconds]
- Frame Number: 29
- Frame Length: 98 bytes (784 bits)
- Capture Length: 98 bytes (784 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: ethiethertype:ip:icmp:data]
- [Coloring Rule Name: ICMP]
- [Coloring Rule String: icmp || icmpv6]
- Ethernet II, Src: Private 66:68:00 (00:50:79:66:68:00), Dst: 0c:41:66:42:00:00 (0c:41:66:42:00:00)
 - Destination: 0c:41:66:42:00:00 (0c:41:66:42:00:00)
 - Source: Private 66:68:00 (00:50:79:66:68:00)
 - Type: IPv4 (0x0800)
 - [Stream index: 4]
- Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1
 - 0100 = Version: 4
 - 0101 ... = Header Length: 20 bytes (5)

Раздел 4

4. Выводы

4.1 слайд 1

Я построил простейшие модели сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, проанализировал трафик посредством Wireshark.